Jane Doe

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Projects

Synchronous Buck Converter for a CubeSat Power Board Fall 25 - Spring 26

KiCad, LTspice, STM32, Embedded C, bench testing with an electronic load

- Reached [X]% peak efficiency at a [X] A load, measured on an electronic load, by replacing the catch diode with a synchronous MOSFET stage and tuning dead time to [X] ns with an STM32 timer, up from [X]% before
- Cut output ripple from [X] mV to [X] mV peak to peak by redesigning the [X]-layer PCB in KiCad with a solid ground plane and a tighter switching loop, then resizing the LC output filter for [X] kHz switching
- Wrote STM32 firmware in C that samples input current at [X] kHz and shuts the rail down within [X] µs of an overcurrent fault, which kept the board alive through [X] deliberate short-circuit tests at a full [X] V input
- Kept the FETs under [X] °C at full load by adding thermal vias and a 2 oz copper pour under them

FPGA Five-Band Audio Equalizer Spring 25

Verilog, Vivado, MATLAB, Basys 3 FPGA, fixed-point DSP

- Processed [X] kHz stereo audio at [X.X] ms latency by pipelining [X] [X]-tap FIR filters in Verilog
- Matched the MATLAB response within [X.X] dB from [X] Hz to [X] kHz by testing coefficients in Vivado
- Cut LUT use [X]% by sharing [X] multiplier per filter across all [X] taps with time-division scheduling

Bluetooth Battery Monitor Fall 24

ESP32, Arduino C++, Python, INA219 current sensor

- Tracked [X] Li-ion cells within [X] mV each with an ESP32 monitor that streams to a Python dashboard
- Extended pack runtime [X]% in field tests by adding a cutoff that drops non-critical loads at [X.X] V

Experience

Power Subsystem Engineer, State University CubeSat Team Fall 24 - Spring 26

- Cut power board mass by [X] g by replacing [X] linear regulators with [X] buck converter board, which then passed thermal vacuum testing from -[X] °C to [X] °C on the first attempt with no board rework needed
- Found [X] battery protection faults before launch review by writing a Python test script that runs [X] charge and discharge cycles overnight and flags any cell that drifts more than [X] mV from the pack average

Undergraduate Research Assistant, Power Electronics Lab, State University Summer 25

- Reduced GaN switching loss measurement error from [X]% to [X]% by adding probe deskew calibration to the double-pulse test setup, which the lab now uses as the standard setup for every GaN device it tests
- Cut test time per device from [X] to [X] minutes by automating oscilloscope capture with Python and PyVISA

Education

B.S. Electrical Engineering, GPA: [X.XX]/4.00 Expected Spring 27

State University

Relevant coursework: Signals and Systems, Electronics I and II, Power Electronics, Digital Design, Embedded Systems, Control Systems, Electromagnetics, Probability for Engineers, Microcontroller Lab

Activities: Tau Beta Pi, IEEE student branch

Skills

Certifications: CSWA (Certified SolidWorks Associate), IPC J-STD-001 soldering

Software: KiCad, LTspice, MATLAB, Vivado, Python, C, Verilog, SolidWorks, Git